

CollectiveFL: Edge-to-Edge Collective Intelligence Transfer in Federated Continual Learning

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Abstract— With the rise of intelligent services on edge devices, the focus of intelligence formation has shifted to the user-side, enabling faster and customized services near the data source, without network overhead or privacy concerns. However, on-device edge intelligence faces some challenges caused by limited data availability and resource constraints in computation and memory. Fortunately, there is more and more intelligence nearby. To effectively harness the potential of widespread edge intelligence, we introduce *CollectiveFL*, a federated edge intelligence framework that facilitates de-biased, robust edge-to-edge knowledge transfer. By tailoring knowledge sharing for each device based on decision logic similarity, we ensure that edge-side learners specialize in their respective purposes and leverage purpose-specific data. Importantly, to mitigate some possible biases on their own or transferred local data, we delegate knowledge transfer to a set of selected neighboring devices rather than one. By sharing and consolidating collective yet customized intelligence, *CollectiveFL* establishes collaborative edge-only intelligence without the help of remote servers. Our extensive experimental results on four different model architectures using 13 public datasets have demonstrated that *CollectiveFL* enhances local learning in 55 out of 63 cases (87.3%) while improving the accuracy of individual tasks by up to 14.1%.

Index Terms—Federated Continual Learning, Collective Intelligence, Edge Intelligence, Edge-to-Edge Knowledge Transfer

I. INTRODUCTION

Edge Artificial Intelligence (AI) has emerged as a promising approach to tackle the cloud-based centralized AI, in terms of communication bottlenecks, unacceptable latency, unavailability under network disconnections or server failures, and privacy concerns [1], [2]. The increasing demand for personalized intelligent services, tailored to individual user requirements, has further placed a heavy burden on central cloud servers. In response to these challenges, on-device, edge intelligence has emerged as a practical solution while offering more personalized intelligent services directly at the edge [3]. Although edge intelligence holds great promise, it still faces a fundamental challenge due to the constrained resources or locally biased data on edge devices.

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While cloud-based AI relies on one (or a few) powerful centralized resources, edge-side AI has distributed resources and data spread across multiple devices, each with limited resources but collectively offering a larger resource and diverse data. Motivated by widespread edge intelligence and success in transfer learning [4], [5], we can empower edge AI learners to achieve broader knowledge by facilitating edge-to-edge knowledge transfer and exposing learners to indirect experiences. Here, each edge device serves various purposes and possesses distinct and specialized characteristics, aiming for personalized intelligence. However, typical knowledge reuse approaches often focus on having generally good knowledge [6]. Popular knowledge reuse methods using ImageNet-pretrained models [7] have made significant advancements in numerous intelligent tasks. However, these methods fail in providing task-specific, personalized assistance. Furthermore, they are designed only for a large-scale architecture trained on vast amounts of powerful data, making them impractical and infeasible at resource-constrained edge devices.

To address these challenges, we present to enhance the capabilities of edge-only intelligence and foster a collaborative yet personalized system. We pose three key questions:

(i) *How to fully harness the potential of widely spread edge intelligence?* – Given the unique understanding of private objectives in widespread edge intelligence, we facilitate selective edge-to-edge knowledge transfer. However, when reusing other’s knowledge, we often suffer from the catastrophic forgetting problem in heterogeneous settings, which is to forget the previously trained knowledge and mix it with the newly trained knowledge. To handle the heterogeneity across devices, we adopt a continual learning paradigm when reusing the knowledge—preserving it to get broader knowledge while isolating task-specific knowledge to avoid interference.

(ii) *How to discover personalized, beneficial knowledge for an edge device with its own specific tasks?* – If some external knowledge exhibits similar decision logic patterns to the learner’s task, it can provide task-specific desirable knowledge. Since external knowledge is extracted from various distinct semantic understandings, it can provide some more generalized and broader understanding that a learner cannot attain only from its local data. This can enable successful extension for unseen data by preventing the overfitting problem.

(iii) *How to construct robust knowledge against locally*

biased edge intelligence? – In edge networks, there may exist some biased knowledge which is beneficial for certain tasks but can be negative for others [6]. To mitigate potential risks of adversely affecting the local learner, we delegate the assistance to multiple models, rather than relying on a single knowledge. We design to aggregate collective intelligence as a backbone to safeguard against adverse effects. Leveraging the collective, personalized, and de-biased intelligence across devices fosters a broader and more diverse understanding of data patterns, ultimately leading to improved performance and generalization across various tasks and datasets.

Based on these premises, we propose *CollectiveFL*, a federated continual learning framework for empowering private tasks via selective, customized, and de-biased edge-to-edge collective knowledge transfer. By probing neighboring knowledge, we identify a set of models with desirable generalized characteristics by assessing the similarity in decision patterns. We then combine multiple models to form a collective backbone of knowledge by adopting some regularization approaches in federated learning [8]. Subsequently, individual learners can effectively train their private tasks on top of the collective knowledge in a continual manner.

The main contributions can be summarized as follows:

- We introduce a novel federated continual learning framework, empowering edge devices to become collaborative yet personalized edge-only intelligence, reducing the reliance on a centralized agent.
- We facilitate de-biased collective knowledge transfer, mitigating the potential of locally biased knowledge that could adversely degrade the quality of target knowledge.
- Extensive experiments on 13 datasets have demonstrated that *CollectiveFL* consistently leads to stable improvements in 55 out of 63 cases (87.3%) in performance, with accuracy improvements of up to 14.1%.

II. RELATED WORK

Federated learning [9], [10], where multiple devices collaborate to achieve better knowledge, has been widely investigated to support edge intelligence. However, federated learning typically processes one global common goal based on local data, limiting its suitability for various edge intelligence scenarios. Although some personalized federated learning methods [11], [12] have been explored to adapt to heterogeneous local data, they still focus on optimizing for a single global knowledge.

Several approaches of knowledge reuse and transfer learning have been applied [13], [14]. Some common transfer learning methods use ImageNet-pretrained models [7], but they provide only a few large-scale architectures such as ResNet [15] or VGG [16], which are not suitable for resource-constrained edge devices. Additionally, each private task requires different knowledge, making unified knowledge unsuitable for providing customized help [6]. With the proliferation of diverse edge intelligence, edge-to-edge knowledge transfer [13], [17] emerges as a promising approach to enhance local private tasks. Furthermore, to overcome the catastrophic forgetting

problem in transfer learning, task-incremental continual learning methods [18]–[20] can retain the ability of pre-trained tasks, while accommodating new tasks.

More closely related to our work, federated continual learning has been explored recently [6], [18], [21], [22]. In edge networks, data and optimization at each device exhibit heterogeneity, and individual learners in a federated continual manner can benefit from other intelligence, even with this heterogeneity. For example, *FedWeIT* [6] identifies positive and negative knowledge for a specific target task and introduces masking on the global model based on personal objectives. *FCCL* [21] primarily focuses on the heterogeneity of models and data in multi-task federated learning scenarios. While both approaches offer better fidelity to private tasks by learning generalization from other federated learners, they assume a centralized federated learning framework, still suffering from communication and computation bottlenecks. Our recent work, *Pick-a-back* [22], considers decentralized knowledge transfer; however, when the transferred knowledge is biased, it can often result in an accuracy drop. In contrast, *CollectiveFL* leverages collective, de-biased knowledge transfer, effectively mitigating negative transfer and enhancing overall robustness.

III. SYSTEM DESIGN

Given L independent learners in edge networks, each learner denoted as $l \in \{1, 2, \dots, L\}$ processes its own sequential tasks $T^{(l)} = \{T_1^{(l)}, T_2^{(l)}, \dots, T_t^{(l)}, \dots\}$ using private input data $X^{(l)}$ labeled with $Y^{(l)}$, where $T_t^{(l)} = \{(X_t^{(l)}, Y_t^{(l)})\}$ represents the t -th task at learner l . To address the limitation of catastrophic forgetting, we let edge learners to be task-incremental continual learners, where model parameters $\theta_t^{(l)}$ are trained with the task $T_t^{(l)}$ on its previous model $\theta_{t-1}^{(l)}$ as:

$$\theta_t^{(l)} \leftarrow \min_{\theta} \mathcal{L}(Y_t^{(l)}, \phi(\theta_{t-1}^{(l)}; X_t^{(l)})), \quad (1)$$

where $\mathcal{L}$ is the loss function between the true label Y and the predicted output ϕ . In a distributed manner, each edge learner acts as a master node, performing its continual tasks.

The objective is to enhance the accuracy of a private task via model sharing. Different from traditional continual learning approaches, the learners in federated continual learning can retrieve external knowledge $\hat{\theta}_{t-1}$, as formulated below:

$$\theta_t^{(l)} \leftarrow \min_{\theta} \mathcal{L}(Y_t^{(l)}, \phi(\hat{\theta}_{t-1}; X_t^{(l)})), \quad (2)$$

where the original backbone model $\theta_{t-1}^{(l)}$ in Eq. (1) is replaced by $\hat{\theta}_{t-1}$, which is a collective knowledge selectively chosen from the learner's own neighboring devices (belonging to set $S^{(l)}$). There is no restriction on heterogeneity in terms of tasks and data distribution, but we are limited to a homogeneous model architecture to facilitate knowledge transfer. It means that the dimension of model parameters θ_t remains identical across learners, as observed in typical federated learning.

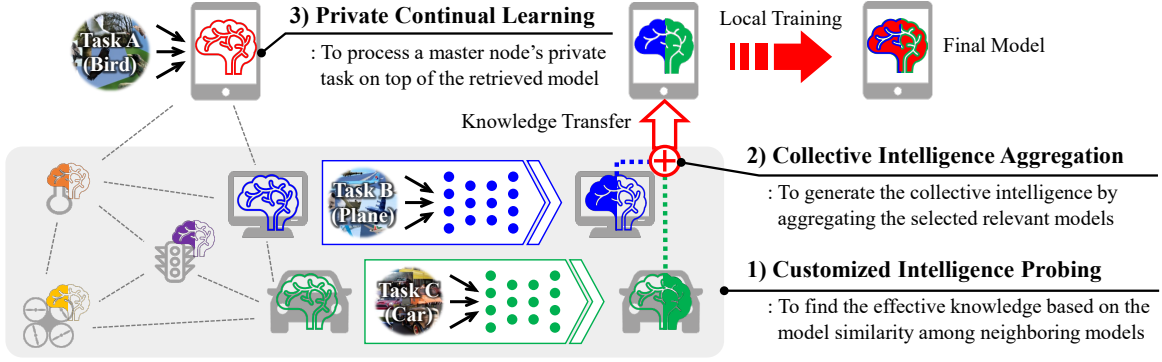


Fig. 1: System overview of *CollectiveFL* for de-biased, collective knowledge transfer in federated continual learning

A. System Architecture

Our approach comprises three core steps (Fig. 1): 1) neighbor knowledge probing and selection for customized intelligence; 2) model aggregation for collective intelligence; and 3) private continual learning. During the neighbor probing step, a master device identifies some neighboring models that can be constructively helpful once tailored to its own private target task, and then multiple neighboring models are merged into a single model using model regularization techniques in the model aggregation step. Subsequently, private continual learning retrieves the aggregated model, and the refined knowledge extracted from a mixture of both knowledge serves as the basis for a master learner to train the private task.

1) *Customized intelligence probing*: To enhance the private task’s capability, each learner probes neighboring knowledge. We quantify the similarity between the master’s local model and its neighbors’ local models. If some external models show similar decision boundary patterns to their internal local models, it indicates that neighboring knowledge can provide a desirable classification capability for its target task. Furthermore, external knowledge has a similar but distinct decision boundary, which enables the master learner to develop more generalized knowledge upon developing its task. This indirect experience empowers the private learner to attain improved performance through customized intelligence selection.

2) *Collective intelligence aggregation*: Even with broader experiences, there exists a potential risk of selecting negative knowledge that could adversely impact the target task. To counteract some possible distortion from the backbone knowledge, we pursue a balance between its own locally trained original knowledge and somewhat broader and generalized knowledge. Therefore, we focus on combining several selective pieces of knowledge based on model similarity in the probing stage. We specifically limit the selection to two different models to achieve broader generalization, while alleviating computation or communication overhead, and the dilution problem caused by excessive regularization. By adopting a simple yet efficient model regularization approach [8], commonly used in federated learning frameworks, we aggregate multiple models into a single model, which serves as the backbone model for the target task.

3) *Private continual learning*: The private continual learning step is responsible for training the target task with the assistance of the aggregated backbone model. In contrast to a typical transfer learning paradigm that entirely replaces its own prior tasks, our framework adopts a continual learning approach to mitigate the issue of catastrophic forgetting and retain the ability of the fetched knowledge. In particular, when our learning approach allows to continue to learn the private task on top of the backbone model, it further extracts some effective neurons from the aggregated model. These steps are repeated for every target task in a sequence.

IV. CUSTOMIZED COLLECTIVE INTELLIGENCE

In typical continual learning, when an edge learner l trains the t -th task $T_t^{(l)} = \{(X_t^{(l)}, Y_t^{(l)})\}$, it constructs model $\theta_t^{(l)}$ without forgetting what it has learned from previous tasks $\{T_1^{(l)}, T_2^{(l)}, \dots, T_{t-1}^{(l)}\}$. To improve performance on the private task $T_t^{(l)}$, we propose to substitute the current model $\theta_{t-1}^{(l)}$ with some other useful external knowledge $\hat{\theta}_{t-1}$ such that:

$$\mathcal{L}(Y_t^{(l)}, \phi(\hat{\theta}_{t-1}; X_t^{(l)})) < \mathcal{L}(Y_t^{(l)}, \phi(\theta_{t-1}^{(l)}; X_t^{(l)})), \quad (3)$$

where $\hat{\theta}_{t-1}$ represents a collective model derived from constructively relevant neighboring knowledge. To generate an effective backbone model, we explore nearby knowledge within its local edge networks. In particular, we inherit some models with similar decision logic, which can offer desirable generalized knowledge, thereby providing valuable indirect experiences. Additionally, to mitigate the risk of harming the target task, we pursue a balance between its own locally trained original knowledge and somewhat broader and generalized knowledge opt for two different models, striking a balance that avoids excessive dilution of the original knowledge. We then aggregate these two models to obtain a broader and more stabilized knowledge base. The refined knowledge extracted from a mixture of both knowledge serves as the basis for a master learner to train in the private task.

A. Customized Intelligence Probing

Different (or heterogeneous) tasks assigned to individual learners at edge devices may lead to quite varied knowledge, resulting in broader or deeper experiences for others. By introducing external knowledge (learned by others) to a local

Algorithm 1 Customized Collective Intelligence for Federated Continual Learning in *CollectiveFL*

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1: Input:  $L$ : # of devices,  $K$ : # of models to aggregate, (for each device  $l$ )  $\theta_0^{(l)}$ : an initial model,  $T^{(l)}$ : a sequence of tasks
2: for task  $T_t^{(l)} \in \{1, 2, \dots\}$  do
3:   for neighbor device  $i = 1 : L$  do ▷ Intelligence Probing
4:     Generate pair-wise adversarial set  $\{(X'_{norm}, X'_{adv})\}$ ;
5:      $DDV_{\theta_t^{(i)}} = \text{dist}(\phi(\theta_t^{(l)}; X'_{norm}), \phi(\theta_t^{(i)}; X'_{adv}))$ ;
6:      $DDV_{\theta_{t-1}^{(i)}} = \text{dist}(\phi(\theta_{t-1}^{(l)}; X'_{norm}), \phi(\theta_{t-1}^{(i)}; X'_{adv}))$ ;
7:      $\text{sim}(\theta_t^{(l)}, \theta_{t-1}^{(i)}) = \text{cosine\_similarity}(DDV_{\theta_t^{(l)}}, DDV_{\theta_{t-1}^{(i)}})$ ;
8:   end for
9:    $S^{(l)} = \text{argmax}_i \text{sim}(\theta_t^{(l)}, \theta_{t-1}^{(i)})$ ; ▷ Intelligence Aggregation
10:   $\bar{\theta}_{t-1}^{(l)} = \text{regularization}(\theta_{t-1}^{(s_1)}, \theta_{t-1}^{(s_2)}, \dots, \theta_{t-1}^{(s_K)})$ ;
11:   $\hat{\theta}_{t-1}^{(l)} \leftarrow \text{masking}(\bar{\theta}_{t-1}^{(l)})$ ; ▷ Private Continual Learning
12:  Freeze  $\hat{\theta}_{t-1}^{(l)}$  and expand a new space for the target task  $T_t^{(l)}$ ;
13:  Train task  $T_t^{(l)}$ ,  $\theta_t^{(l)} \leftarrow \mathcal{L}(Y_t^{(l)}, \phi(\hat{\theta}_{t-1}^{(l)}; X_t^{(l)}))$ ;
14: end for

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learner, the model gains the ability to leverage a wider range of feature representations, leading to a broader understanding of patterns and information that can be advantageous for its specific learning objective. As the external models have learned alternative representations that capture specialized knowledge relevant to specific tasks, strategically transferring customized knowledge may be an effective approach for enhancing performance at the edge device.

Given that a master learner l has its neighboring models $\theta_{t-1}^{(i)}$ for $i \in \{1, 2, \dots, L\}$, we evaluate model similarity between its local model and the neighbors' models. We adopt the similarity metric, *ModelDiff* [23] that measures the decision boundary similarity between two different models. The base model for the master learner is $\theta_t^{(l)}$, where the initial model $\theta_1^{(l)}$ is a scratch model trained solely on the first task $T_1^{(l)}$. We compute the similarity metric between the master model $\theta_t^{(l)}$ and another model $\theta_{t-1}^{(i)}$ of neighbor device i as follows:

$$\text{sim}(\theta_t^{(l)}, \theta_{t-1}^{(i)}) = \text{cosine_sim}(DDV_{\theta_t^{(l)}}, DDV_{\theta_{t-1}^{(i)}}). \quad (4)$$

DDV_{θ} denotes the decision distance vector with model θ . DDV is computed by injecting some pair-wise adversarial input sets into θ and then taking the Euclidean distance as:

$$DDV_{\theta} = \text{dist}(\phi(\theta; X'_{norm}), \phi(\theta; X'_{adv})), \quad (5)$$

where X'_{norm} and X'_{adv} are pair-wise sets of normal samples and adversarial samples, derived from the data of $\{T_t^{(l)} \cup T_{t-1}^{(i)}\}$ to effectively identify the decision logic [23], [24]. We sequentially assess the similarity between the distance vectors of the master model $\theta_t^{(l)}$ and each candidate model $\theta_{t-1}^{(i)}$ across all possible neighboring models.

B. Collective Intelligence Aggregation

After acquiring the similarity with each neighboring model, we choose the edge learner set $S^{(l)} = \{s_1, s_2, \dots, s_K\}$ for

master learner l with the highest similarity values as:

$$\text{argmax}_i \text{sim}(\theta_t^{(l)}, \theta_{t-1}^{(i)}). \quad (6)$$

argmax_i is used to select K neighboring models among candidates. Here, the learner's previous model $\theta_{t-1}^{(l)}$ is also included as one of the candidate models.

We now aggregate the neighboring knowledge selected from K models. The diverse knowledge has been trained on distinct datasets, capturing multiple perspectives and feature extraction capabilities. By drawing upon a diverse ensemble of knowledge sources, the learner can benefit from their complementary expertise, leading to improved accuracy and robustness.

We exploit the existing model regularization approach such as *FedAVG* [8] to aggregate them as follows:

$$\bar{\theta}_{t-1}^{(l)} = \text{regularization}(\theta_{t-1}^{(s_1)}, \theta_{t-1}^{(s_2)}, \dots, \theta_{t-1}^{(s_n)}), \quad (7)$$

where n represents the number of selected candidates $S^{(l)}$. We mainly use $n = 2$, since a larger n may lead to excessive computation/communication costs and a side effect of diluting the original knowledge. Further validation will be in Sec. V.

C. Private Continual Learning

Finally, the private learner proceeds to train the private task $X_t^{(l)}$ using the aggregated model as a base model. It is important to note that our approach is independent of any continual learning frameworks that regularly expand the model architecture [20], [25]. Different from typical continual learning methods that only concatenate previous knowledge, we retrieve useful information from the model, inspired by *ProgressiveNet* [25] and *CPG* [20], [25]. This facilitates more solid generalization to unseen examples and aids in mitigating the overfitting problem, resulting in improved performance.

We extract essential partial models from the external knowledge. To extract effective neurons from the aggregated model $\bar{\theta}_{t-1}^{(l)}$, we conduct additional masking by borrowing an idea from *Piggyback* [20], [26]. A pruned model $\hat{\theta}_{t-1}^{(l)}$ is loaded as a backbone model $\theta_{t-1}^{(l)}$ for master learner l as follows:

$$\theta_{t-1}^{(l)} \leftarrow \hat{\theta}_{t-1}^{(l)} = \text{masking}(\bar{\theta}_{t-1}^{(l)}). \quad (8)$$

After freezing the backbone model to prevent the catastrophic forgetting problem, we expand the model for the target task with random initialization and then train the task, while optimizing the following loss:

$$\theta_t^{(l)} \leftarrow \mathcal{L}(Y_t^{(l)}, \phi(\theta_{t-1}^{(l)}; X_t^{(l)})), \quad (9)$$

where a set of $X_t^{(l)}$ and $Y_t^{(l)}$ is a private task.

The detailed procedure is described in Algorithm 1.

V. EXPERIMENTS

Datasets and task specifications. We used 13 public datasets and organized them into 4 distinct groups, as summarized in

¹As the image size of 32×32 is not feasible on EfficientNet-B0, we upscale the images of CIFAR-100 to 224×224 .

²Arabia (MNIST-9) is the same dataset with Arabia (MNIST-10) that removes the class of digit 0.

TABLE I: Summary of datasets (‘*NID*’ indicates that the size of images within the dataset is not identical)

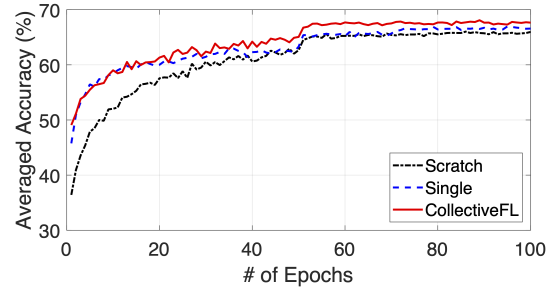
Group	Dataset	# of Tasks (= # of Superclasses)	# of Subclasses per Task	Original Image Size	Resized Image Size	# of Training Data	# of Test Data
	CIFAR-100	20	5	32×32	32×32^1	500×100	100×100
	Classified-ImageNet	31	10	-	224×224	422,849	16,500
Fine-Grained	CUB		200			5,994	5,794
	Stanford Cars		196			8,144	8,041
	Flower	5	102	<i>NID</i>	224×224	2,040	6,149
	WikiArt		195			42,129	10,628
	Sketch		250			16,000	4,000
Digits	Arabia (MNIST-10) ²			28×28		33,601	8,399
	Arabia (Synthetic)		10 (0~9)	<i>NID</i>		10,000	2,000
	Arabia (SVHN)			32×32		73,257	26,032
	Arabia (MNIST-9) ²	7		28×28	32×32	33,601	8,399
	Arabic		9 (1~9)	28×28		54,000	9,000
	Chinese			64×64		7,200	1,800
	Roman			<i>NID</i>		1,888	472

Table I. Each device is assigned a completely heterogeneous task group by partitioning the classes without any overlap. (i) For *CIFAR-100*, the original dataset is provided in 20 superclasses (i.e., device) with 5 classes per task, with a total of 100 classes [27]. (ii) In the case of *Classified-ImageNet* [28], we divided 310 classes into 31 tasks, with 10 classes each, based on the word hierarchy provided by the English lexical database, *WordNet* [29]. (iii) The *Fine-Grained* (consisting of CUB [30], Stanford Cars [31], Oxford Flowers [32], WikiArt [33], and Sketch [34]) and (iv) *Digits* (consisting of Arabia (MNIST-10) [35], Arabia (Synthetic) [36], Arabia (SVHN) [37], Arabia (MNIST-9) [35], Arabic [38], Chinese [39], and Roman [40]) datasets were obtained from completely different sources, featuring a varying number of classes per task.

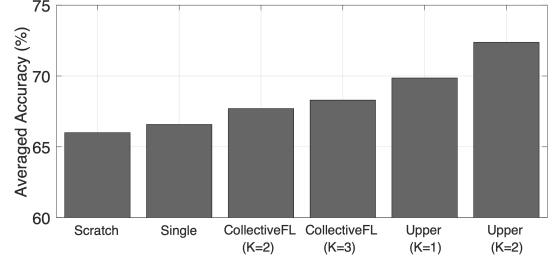
Model configurations. We implemented a range of architectures, from lightweight models suitable for edge devices to large-scale models with stronger performance, which may not be feasible for edge deployment. The models utilized include LeNet-5 [41], MobileNetV1 [42], EfficientNet-B0 [43], and VGG16 with Batch Normalization (VGG16-BN) [16]. Our primary architecture is LeNet-5, a classic and straightforward model. It should be noted that although there are further variations of LeNet-5 with improved performance using dropouts or different activation functions, we used the most naive vanilla LeNet-5 with Tanh activation without additional modifications.

Learning parameters. Each model was trained for 100 epochs with a batch size of 32. We used an initial learning rate of 0.01, which was decreased by a factor of 0.1, at the epochs of 50 and 80. The Stochastic Gradient Descent (SGD) with a momentum of 0.9 was used as the default optimizer. We adhere to the original settings of activation functions, dropout rates, and importantly, assigned learning parameters.

Baselines. In our experiments, the selection of $K = 2$ turns out to be effective, where communication and computation costs can also be minimized. We compared our *CollectiveFL* with the following approaches: 1) *Scratch*: a naive model that does not use any external knowledge; 2) *Upper*: $K = 2$ but



(a) Training dynamics



(b) Overall accuracy

Fig. 2: Averaged accuracy on 20 tasks

with an ideal optimal solution by training on all combinations in a brute-force manner; 3) *Weighted AVG*: employing a different regularization method that uses the similarity measure as the weight for the weighted averaging; and 4) *ImageNet-Pretrained*: using the ImageNet-pretrained models provided by *PyTorch* [7] as a backbone knowledge in large-scale models such as VGG16-BN or EfficientNet-B0. We also compared *CollectiveFL* with existing (federated) continual learning approaches using our backbone selection with $K = 1$ as follows: 5) *Single (Pick-a-back* [22]): federated continual learning with knowledge transfer, while selecting a single model as the backbone model; 6) *FedWeIT* [6]: federated continual learning with task-dependent masking on a global model; 7) *WSN* [44]: continual learning to solve catastrophic forgetting problem; and 8) *FCCL* [21]: federated continual learning to solve the catastrophic forgetting problem.

Metrics. We primarily report the average test accuracy (Avg), along with the highest improvement (Top) achieved through collective knowledge transfer. To demonstrate the positive influence of *CollectiveFL*, we provide the number and ratio of positive and negative cases where the backbone knowledge either assists or harms the local task. Additionally, we provide the average training time per epoch, which was calculated over 5 epochs of 20 tasks. The model size is measured based on the actual space used to store the model.

A. Effect of Customized Collective Intelligence

We investigated the effect of collective intelligence in four sets, compared with naive learning, other aggregation methods, single knowledge transfer, and ImageNet-pretrained models.

Training dynamics. We analyzed the averaged accuracy of 20 tasks in CIFAR-100 using LeNet-5. As shown in Fig. 2, we observed that *CollectiveFL* and *Single* exhibit superior performance from the early stages of training, suggesting that edge-

TABLE II: Accuracy of each task per device sorted by scratch accuracy

Task	15	17	1	16	18	12	3	13	8	14	7	6	9	2	4	19	10	5	20	11	Avg
Scratch	38.8	52.0	56.4	57.0	59.6	63.2	65.2	65.4	66.8	67.2	67.6	68.4	69.2	71.0	72.2	72.8	74.4	74.6	78.2	79.4	66.0
Single	39.4	53.2	57.8	58.0	60.2	65.2	63.2	72.6	70.1	66.8	65.0	67.2	70.2	73.4	69.8	71.0	72.4	74.4	80.0	80.8	66.6
FedWeIT	35.2	49.8	50.0	57.7	58.0	57.7	57.3	62.5	59.3	59.7	65.2	64.0	53.3	64.7	63.7	62.0	67.2	66.5	66.3	80.0	60.0
WSN	31.6	49.8	54.8	56.8	56.4	60.0	63.0	69.2	62.2	61.6	64.0	65.0	56.8	66.8	66.6	63.6	67.0	67.6	73.2	78.8	61.7
FCCL	33.8	49.4	55.4	58.6	59.6	61.0	62.8	66.2	69.0	63.0	68.4	64.0	68.4	68.0	70.4	69.4	73.2	71.6	76.0	77.2	64.3
CollectiveFL	37.4	54.4	57.8	57.6	64.4	65.4	66.2	72.2	72.6	68.0	68.6	68.6	71.4	72.0	71.8	69.0	76.8	75.6	82.0	81.4	67.7
	(-1.4)	(+2.4)	(+1.4)	(+0.6)	(+4.8)	(+2.2)	(+1.0)	(+6.8)	(+5.8)	(+0.8)	(+1.0)	(+0.2)	(+2.2)	(+1.0)	(-0.4)	(-3.8)	(+2.4)	(+1.0)	(+3.8)	(+2.0)	(+1.7)

to-edge knowledge transfer becomes effective quickly for the target task. This rapid increase in accuracy from the beginning of training accelerates the overall training process and reduces the training time. When comparing *CollectiveFL* with *Single*, *CollectiveFL* consistently outperforms *Single* throughout the training process. This means that broader yet well-selected knowledge transfer benefits the learning process.

Comparison to baselines. The final accuracy compared to baselines is summarized in Fig. 2(b). We observe a steady increase in accuracy, progressing from models without knowledge transfer through single knowledge transfer and finally to multiple knowledge transfer. *Upper* as the ideal case achieved the best performance through evaluation of all combinations of selecting one or two tasks out of the 20 available tasks in a brute-force manner. It demonstrates a remarkable improvement, reaching 72.4% with an average improvement of 6.4%, indicating that retrieving multiple external knowledge significantly benefits a stand-alone learner. Although *CollectiveFL* cannot achieve *Upper*, *CollectiveFL* still provides valuable assistance. We compared *CollectiveFL* with several (federated) continual learning approaches in Table II.³ The federated continual learning with collective intelligence in *CollectiveFL* outperforms existing (federated) continual learning approaches with a gap of 7.7% to 3.4%. Notably, *CollectiveFL* shows the best accuracy in 18 tasks among 20 cases.

Impact across various datasets. In Table III, we presented the performance across different datasets: from Digits of a relatively simple task, CIFAR-100, Classified-ImageNet, and to Fine-Grained of a very complicated task. By leveraging customized knowledge, either from single or multiple sources, we observe positive enhancements in 5 out of 7 cases, 17 out of 20 cases, 29 out of 31 cases, and 4 out of 5 cases, respectively, resulting in an overall improvement rate of 87.3% (55 out of 63 cases) with *CollectiveFL*, and 76% (48 out of 63 cases) with *Single*. Furthermore, by mitigating the risk of biased knowledge through multiple neighbor model selection, the negative cases have been reduced from 3, 8, 3, and 1 (totaling 15) in *Single* to 1, 3, 2, and 1 (totaling 7) in *CollectiveFL*.

CollectiveFL has achieved 3.0%, 6.8%, 11.0%, and 14.1% accuracy improvement at best, and 0.9%, 1.7%, 4.2%, and 4.7% on average in Digits, CIFAR-100, Classified-ImageNet, and Fine-Grained, respectively. The results show that there is a higher improvement in both average and best cases when using

TABLE III: Different datasets with LeNet-5

		Scratch	Single	CollectiveFL
Digits	Avg	94.0	94.2	94.9 (+0.9)
	Top	82.4	83.9	85.4 (+3.0)
CIFAR-100	Avg	66.0	66.6	67.7 (+1.7)
	Top	65.4	72.6	72.2 (+6.8)
ImageNet	Avg	40.9	44.1	45.1 (+4.2)
	Top	61.8	65.6	68.6 (+6.8)
Fine-Grained	Avg	15.6	16.9	20.3 (+4.7)
	Top	14.4	17.0	28.5(+14.1)

TABLE IV: Performance of Fine-Grained dataset with EfficientNet-B0

Task	1	3	5	4	2	Avg
Scratch	8.3	7.4	12.7	28.2	66.3	24.6
Pretrained	28.9	42.4	37.1	49.9	90.5	49.8
CollectiveFL	42.4	42.9	60.5	52.5	76.1	54.9
	(+34.1)	(+35.5)	(+47.8)	(+24.3)	(+9.8)	(+30.3)

collective intelligence for complicated tasks. This indicates that more complex tasks with lower accuracy have greater potential to benefit from selective knowledge transfer. In the fine-grained dataset, which consists of 102 to 250 classes per task and is challenging to process using LeNet-5, we observe relatively poor results compared to other datasets. However, even in such challenging cases, *CollectiveFL* still provides significant assistance to all tasks.

Performance of the most complicated task. To further validate our findings from Table III, where LeNet-5 was not sufficient for reasonable training, we conducted additional experiments using EfficientNet-B0 on the Fine-Grained dataset. As shown in Table IV, the scratch accuracy of EfficientNet-B0 is much higher than that of LeNet-5. Remarkably, *CollectiveFL* substantially improves the overall accuracy by 30.3% on average and up to 47.8% at best. Although these results may seem surprising, they are in line with the intention of *ImageNet-Pretrained* approaches. This has also been demonstrated in previous works, such as *CPG* [20] and *Piggyback* [26], where scratch accuracy with ImageNet-pretrained models on ResNet50 and VGG16-BN achieved 58.83% and 43.84%, respectively, on average of five tasks. With the collective intelligence of *CollectiveFL*, our approach not only outperforms the accuracy of ImageNet-pretrained VGG16-BN [26], but also operates comparably to the performance of ImageNet-

³The implementation (including learning parameters) is set on top of open sources from *FedWeIT*, *WSN*, and *FCCL*.

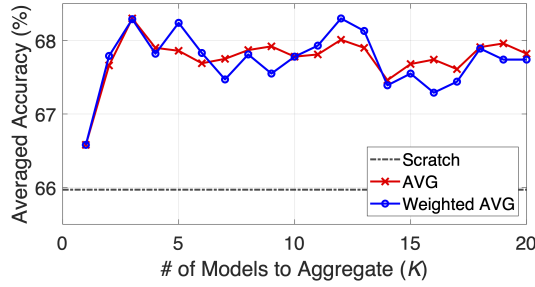


Fig. 3: Accuracy with respect to the different number of models to aggregate compared to naive averaging aggregation

pretrained ResNet50 [20], by using a considerably lightweight model, EfficientNet-B0. This significant achievement showcases the potential of *CollectiveFL* in harnessing the full capabilities of edge devices while maintaining efficiency.

It is worth noting that ImageNet-pretrained models are available only at a limited number of architectures and require an additional 92.5 GB of data for training locally. However, in many cases, they become less effective due to the lack of customization in knowledge selection compared to our framework. On the other hand, *CollectiveFL* wisely leverages edge-to-edge knowledge transfer to provide a customized backbone, outperforming *ImageNet-Pretrained* on average and excelling in almost every single task (except for Task #2). This demonstrates the superiority of *CollectiveFL* in delivering tailored and effective solutions for edge intelligence applications. **Impact of aggregation.** We investigated the impact of different numbers of models to aggregate and various regularization methods in Fig. 3. In all cases, both fair averaging (AVG) and weighted averaging (*Weighted AVG*) show better improvement compared to using a single model (i.e., $K = 1$) and to training from scratch without knowledge reuse. As expected, aggregating knowledge from too many models dilutes the original knowledge, while the usage of $K = 2$ and $K = 3$ proves to be effective. As our default settings, we used *FedAVG*, denoted as *AVG*, with two models to aggregate, effectively balancing computation and communication overhead, which should be crucial considerations at edge environments. Interestingly, when K becomes larger (i.e., more models to be aggregated), *Weighted AVG* slightly outperforms *AVG*. This suggests that our similarity-based weighting may assign more importance to relevant knowledge, leading to better improvement in accuracy.

B. Applicability across Various Settings

We validated *CollectiveFL* across various architectures, learning seeds, different task sets, and further heterogeneous settings. Our results are presented in terms of averaged accuracy and with the most effective case of the highest improvement achieved through collective knowledge transfer. **Diverse model architectures.** In Table V, we validated *CollectiveFL* on CIFAR-100 using a variety of architectures, ranging from lightweight models like LeNet-5, MobileNetV1, and EfficientNet-B0 to larger-scale models such as VGG16-BN. The training time was measured by averaging five trials for 20 tasks, while the model size was evaluated based on

TABLE V: Different architectures for CIFAR-100

		Accuracy			Model Size	Training Time
		Scratch	Single	CollectiveFL		
LeNet-5	Avg	66.0	66.6	67.7 (+1.7)	0.3 MB (1.0 $\times$)	2.48 s (1 $\times$)
	Top	65.4	72.6	72.2 (+6.8)		
Mobile NetV1	Avg	65.4	66.0	66.6 (+1.2)	15.8 MB (50.7 $\times$)	7.43 s (3.0 $\times$)
	Top	59.8	66.2	65.6 (+5.8)		
Efficient Net-B0⁴	Avg	77.8	80.2	75.4 (-2.4)	19.9 MB (63.7 $\times$)	26.60 s (10.7 $\times$)
	Top	61.4	68.2	65.8 (+4.4)		
VGG16 -BN	Avg	78.7	79.6	80.2 (+1.5)	164.2 MB (526.3 $\times$)	6.55 s (2.6 $\times$)
	Top	76.2	77.6	80.8 (+4.6)		

TABLE VI: Different learning seeds

	Seed 1		Seed 2		Seed 3	
	Avg	Top	Avg	Top	Avg	Top
Scratch	66.0	65.4	66.2	67.4	66.0	63.0
Single	66.6	72.6	66.8	70.6	66.3	65.0
CollectiveFL	67.7 (+1.7)	72.2 (+6.8)	68.0 (+1.8)	73.8 (+6.4)	67.8 (+1.8)	70.4 (+7.4)

the actual memory usage compared to LeNet-5 as a reference scale. We observed that *CollectiveFL* outperformed all cases of *Scratch* and *Single* except for *Single* on EfficientNet-B0. Nevertheless, there was still a significant improvement of up to 4.4% in the most effective case, even with EfficientNet-B0, considering that EfficientNet-B0 requires 10.7 $\times$ longer for training. While large-scale architectures can achieve better accuracy, they come with higher demands for storage and processing resources. For example, VGG16-BN takes 526.3 $\times$ more storage compared to LeNet-5. The trade-off between model complexity and computation or storage cost is critical, where the overhead is a severe challenge for edge devices.

Training stability. To assess the stability, we conducted a simple comparison on CIFAR-100 using LeNet-5 with different learning seeds. The training stability of *CollectiveFL* is a promising aspect, as it ensures reliable and consistent performance across different devices. This is particularly important in real-world scenarios where edge devices may have varying resources and initial conditions. As in Table VI, there was almost no gap among the results. This stability enhances its adaptability to diverse edge environments, enabling seamless and efficient knowledge transfer among edge learners.

Various task configuration. In Table VII, we investigated the impact of different superclass (i.e., task) configurations in CIFAR-100 by grouping them into 10 and 5 superclasses, with 10 subclasses and 20 subclasses, respectively, whereas the original CIFAR-100 is provided in the hierarchy of 20 superclasses with 5 subclasses for each. As the number of classes within the task increases, the tasks become more

⁴The performance of EfficientNet-B0 was assessed on the enlarged CIFAR-100 with the size of 224×224 , but the model size is measured with the input size of 32×32 for fair comparison.

TABLE VII: Different task specification

	20 sup $\times$ 5 sub		10 sup $\times$ 10 sub		5 sup $\times$ 20 sub	
	Avg	Top	Avg	Top	Avg	Top
Scratch	66.1	61.8	59.0	56.6	52.4	46.9
Single	66.7	65.5	59.5	61.2	54.7	49.8
CollectiveFL	68.8 (+2.7)	68.6 (+6.8)	61.4 (+2.4)	63.2 (+6.6)	56.2 (+3.8)	51.9 (+5.0)

TABLE VIII: Different level of heterogeneity

	w/o Sampling		Balanced		Heterogeneous ($\alpha = 0.9$)		Heterogeneous ($\alpha = 0.3$)	
	Avg	Top	Avg	Top	Avg	Top	Avg	Top
Scratch	66.0	65.4	54.0	49.4	52.9	49.8	47.6	35.6
Single	66.6	72.6	57.3	55.8	54.5	52.4	48.2	43.0
CollectiveFL	67.7 (+1.7)	72.2 (+6.8)	55.5 (+1.5)	60.8 (+11.4)	55.2 (+2.3)	56.0 (+6.2)	49.1 (+1.5)	41.2 (+5.6)

complex, resulting in lower accuracy. However, *CollectiveFL* demonstrates its ability to benefit from collective knowledge transfer under various variant tasks, with similar average improvements. Taking a closer look at individual tasks, we found only two negative cases out of both 20 tasks in *20 Super $\times$ 5 Sub* and 10 tasks in *10 Super $\times$ 10 Sub*, and it is even reduced to zero in case of 5 tasks in *5 Super $\times$ 20 Sub*, which is the most complicated configuration. This suggests that in difficult tasks with more room for improvement, *CollectiveFL* is more likely to offer positive knowledge transfer. This analysis provides valuable insights into the effectiveness of edge-to-edge knowledge transfer, indicating that *CollectiveFL* can contribute to rescuing from unsatisfactory performance where further improvement may be needed.

Data heterogeneity. Heterogeneity in data distribution poses a harsh challenge in edge intelligence and federated learning. In our settings, we divided the classes without any overlap or used entirely different datasets, leading to completely heterogeneous data. We addressed this issue by adopting the continual learning paradigm. However, to explore the robustness under even more challenging cases, we introduced additional heterogeneity by sampling data from the original dataset based on the Dirichlet distribution [45], [46], making imbalanced class distributions. We conduct experiments for *w/o Sampling* having 500 training data per class and *Balanced* fairly sampling with 250 training data per each class. A lower value of α in *Dir*(α) indicates a larger heterogeneity with varying numbers of data per class, which can consequently lead to poorer performance. As shown in Table VIII, Despite the challenging and heterogeneous data distributions, *CollectiveFL* consistently demonstrated robust performance on local learning in all scenarios. As each device has its own environment-specific data and objectives, the heterogeneity among local data is a timely challenge. The results imply that our *CollectiveFL* framework suggests a promising direction towards constructing edge intelligence in a robust manner.

VI. DISCUSSIONS

Model architecture. We assume a homogeneous model architecture for edge-to-edge knowledge transfer. However, recent advancements on model compression—such as knowledge distillation [47], [48] or pruning [49], [50]—enable heterogeneous model architectures without sacrificing accuracy or knowledge. After or while transferring the knowledge, these compression techniques can be further applied to reduce model size, allowing for adaptation to heterogeneous device constraints.

Network conditions. We do not consider any client (neighbor) selection. Instead, we find the most informative neighbor within the connectable devices. This approach is still effective even in dynamic networks where network conditions are unstable, or devices frequently join or leave as we find the best among them. Currently, we do not consider scenarios where all the neighbors have negative, harmful knowledge. This can be further leveraged by introducing a similarity threshold; if the similarity scores of neighboring devices are all less than the threshold, we may avoid knowledge transfer. However, we believe that this is not an ordinary case, since we have shown that the knowledge transfer (rather than training from scratch) is beneficial in most representative cases.

Privacy concerns. While mitigating privacy leakage on sharing models by aggregating multiple models and thus preventing the exposure of a single model, there can be additional privacy concerns when calculating similarity scores. However, as demonstrated in [22], even when using noise-injected or just random noise images, we can still discover some helpful knowledge (despite often slightly worse knowledge than when using a target-specific image). This is due to our similarity score, which is to measure how a certain image is interpreted across different models; any input image can be used.

VII. CONCLUSION

We have introduced *CollectiveFL*, a novel approach for building edge-oriented intelligence via selectively transferring collective, de-biased knowledge among learners. In edge networks, each device handles unique local tasks with environment-specific data. We enable edge-side learners to gain valuable indirect experiences by considering the similarity in decision logic, while avoiding possible biases by limiting the sources of transferred knowledge. Extensive experiments across diverse datasets and model architectures have validated the effectiveness of providing both customized and collective intelligence to individual learners.

With the growing importance of intelligent services on edge devices, *CollectiveFL* offers an opportunity for self-sustaining edge intelligence. Inspired by human learning, where students in a group with some common interest and purpose can mutually benefit each other, we demonstrate a promising direction of collective and customized learning methodology for edge-side learners. While further improvements with knowledge distillation and few-shot learning could be explored, our proposed primitive showcases the potential of fostering a strong and balanced knowledge ecosystem for edge intelligence.

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